

Functional Requirements :

User Management (6):

01. Users can create accounts with a username, password, email address, and profile picture. Optionally, a bio and location can be included.
02. Usernames must be unique and follow a specific format (e.g., letters, numbers, and underscores).
03. Email addresses must be valid and unique for account creation.
04. Content moderation: Implement a system for users to report abusive content, with clear procedures for reviewing and handling reports.
05. Account verification: Offer an optional account verification process to increase user trust and potentially grant access to exclusive features.
06. Admins can suspend or ban users for violating terms of service

Content Management (7):

07. Users can post text messages (tweets) adhering to a character limit (e.g., 280 characters).
08. Allow threads where users can post a series of connected tweets (e.g., replying to their own tweet).
09. Users can attach photos and videos to their tweets, with limitations on file size and type.
10. Implement editing capabilities for tweets before posting or after a short window of time.

11. Allow users to delete their own tweets.
12. Mark tweets containing sensitive content as NSFW (Not Safe For Work) with appropriate warnings.
13. Enable content moderation tools to flag and remove tweets violating community guidelines.

Following and Followers (4):

14. Users can follow other users by searching usernames or clicking a "follow" button on profiles.
15. Implement a follower approval system where users can choose to approve followers.
16. Users can see a dedicated list of users they follow and their followers.
17. List creation and management: Allow users to create and manage private or public lists consisting of other users they follow, enabling them to categorize and filter content (replacing requirement 20).

Interactions (7):

18. Users can reply to other users' tweets, inheriting part of the original tweet content (quoted tweet).
19. Users can like (or favorite) other users' tweets.
20. Users can retweet other users' tweets (share them with their followers) with or without additional commentary.
21. Allow quote tweets where users can retweet with their own comment.
22. Implement a direct messaging system where users can send private messages to other users who follow them (or follow each other, optional).
23. Enable notifications for mentions (when someone includes your username in their tweet), replies, and likes on your tweets.

24. Allow users to mute or block other users to prevent them from seeing their tweets or contacting them through direct messages.

Search and Explore (3):

25. Users can search for tweets by keywords, hashtags (preceded by #), and usernames.
26. Implement a trending topics section that displays popular hashtags and keywords based on user activity.
27. Allow curating suggested users to follow based on interests and activity.

Non Functional Requirements :

1. **Scalability:** The system should be able to handle a massive number of users and a high volume of tweets without significant performance degradation.
2. **Availability:** Twitter should be accessible to users 24/7 with minimal downtime for maintenance or emergencies.
3. **Performance:** Users should experience fast response times for actions like posting tweets, refreshing feeds, and searching.
4. **Security:** User accounts, data, and privacy must be protected from unauthorized access, breaches, and malicious activities.
5. **Reliability:** The system should function consistently and reliably, with minimal errors or outages.
6. **Maintainability:** The codebase should be well-documented, modular, and easy to maintain and update.
7. **Testability:** The system should be designed to facilitate comprehensive testing to ensure functionality and identify potential issues.
8. **Internationalization:** Twitter should be accessible and usable in multiple languages and regions, considering cultural nuances.
9. **Accessibility:** The platform should be accessible to users with disabilities, adhering to relevant accessibility standards.